

Synoptic LE: Tree Fluxes 2023

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1 Read in and collate all the files in the Processed Data folder

2 Remove bad fluxes and calculate how many there were

```
##Remove any samples with Flags in the final qaqc_flag columns  
final_dat <- all_dat %>%  
  filter(!grepl("TRUE", CH4_rejected_model, ignore.case = TRUE)) %>%  
  filter(!grepl("TRUE", CO2_flux_rejected , ignore.case = TRUE))
```

```
##Removed both CH4 and CO2 obs when CO2 fluxes were negative and when visual checks showed w
```

```

# count before and after filtering
n_total_co2 <- nrow(all_dat)
n_total_ch4 <- nrow(all_dat)

final_dat <- all_dat %>%
  filter(CH4_rejected_model == FALSE | is.na(CH4_rejected_model)) %>%
  filter(CO2_flux_rejected == FALSE | is.na(CO2_flux_rejected))

n_final <- nrow(final_dat)
n_removed_co2 <- sum(all_dat$CO2_flux_rejected == TRUE, na.rm = TRUE)
n_removed_ch4 <- sum(all_dat$CH4_rejected_model == TRUE, na.rm = TRUE)
n_removed_either <- n_total_co2 - n_final

# build summary table
filter_summary <- tibble::tibble(
  Metric = c(
    "Total observations (joined dataset)",
    "CO2 - observations rejected (n)",
    "CO2 - observations rejected (%)",
    "CH4 - observations rejected (n)",
    "CH4 - observations rejected (%)",
    "Removed by both flags simultaneously (n)",
    "Total unique observations removed (n)",
    "Total unique observations removed (%)",
    "Final observations retained (n)"
  ),
  Value = c(
    n_total_co2,
    n_removed_co2,
    round(100 * n_removed_co2 / n_total_co2, 1),
    n_removed_ch4,
    round(100 * n_removed_ch4 / n_total_co2, 1),
    sum(all_dat$CO2_flux_rejected == TRUE &
      all_dat$CH4_rejected_model == TRUE, na.rm = TRUE),
    n_removed_either,
    round(100 * n_removed_either / n_total_co2, 1),
    n_final
  )
)

filter_summary %>%
  kable(

```

```

caption = "Post-Join QA/QC Filtering Summary",
align   = c("l", "r"),
booktabs = TRUE,
format  = "latex"
) %>%
kable_styling(
  latex_options = c("striped", "hold_position"),
  full_width    = FALSE
) %>%
pack_rows("CO2 flags",          2, 3) %>%
pack_rows("CH4 flags",         4, 5) %>%
pack_rows("Combined filtering", 6, 9)

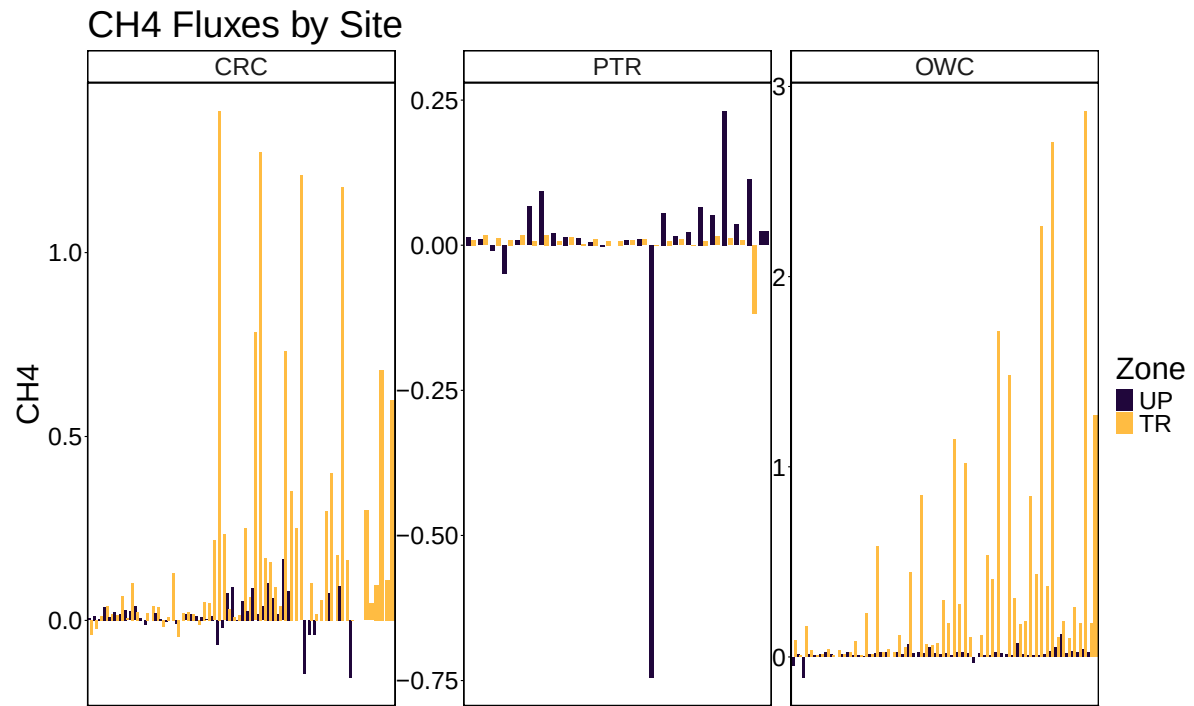
```

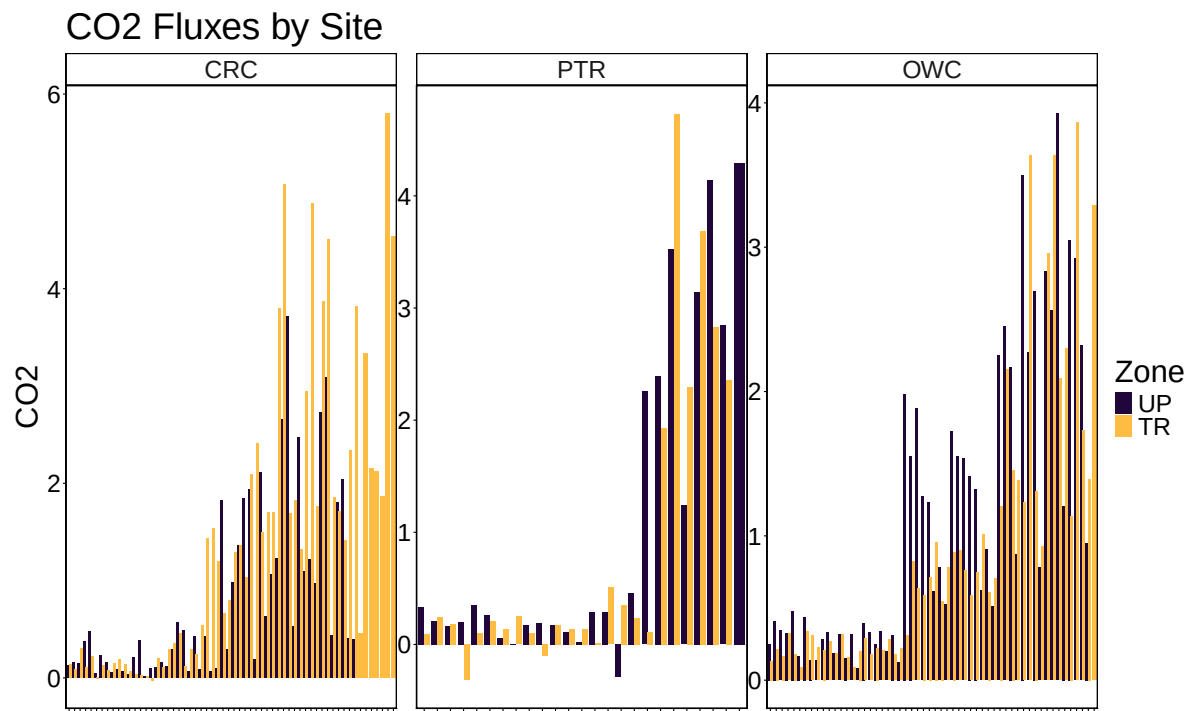
Table 1: Post-Join QA/QC Filtering Summary

Metric	Value
Total observations (joined dataset)	273.0
CO2 flags	
CO2 — observations rejected (n)	7.0
CO2 — observations rejected (%)	2.6
CH4 flags	
CH4 — observations rejected (n)	3.0
CH4 — observations rejected (%)	1.1
Combined filtering	
Removed by both flags simultaneously (n)	2.0
Total unique observations removed (n)	8.0
Total unique observations removed (%)	2.9
Final observations retained (n)	265.0

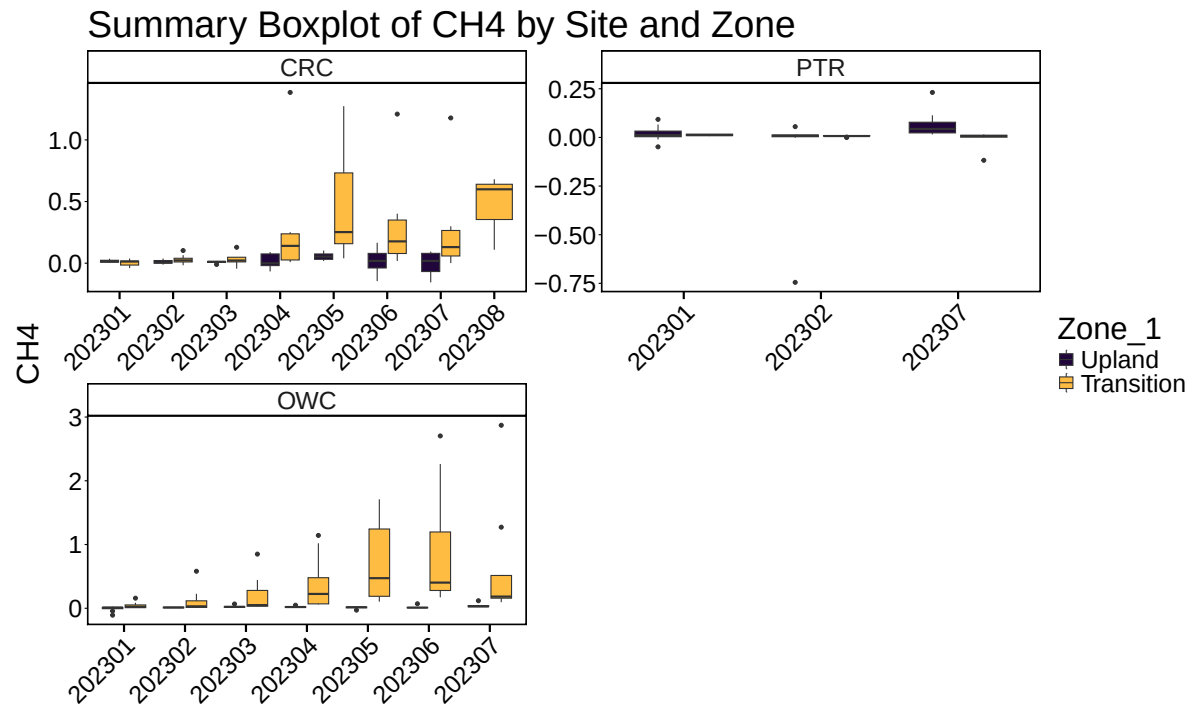
3 Write out files

4 All Data by Year, Site and Zone

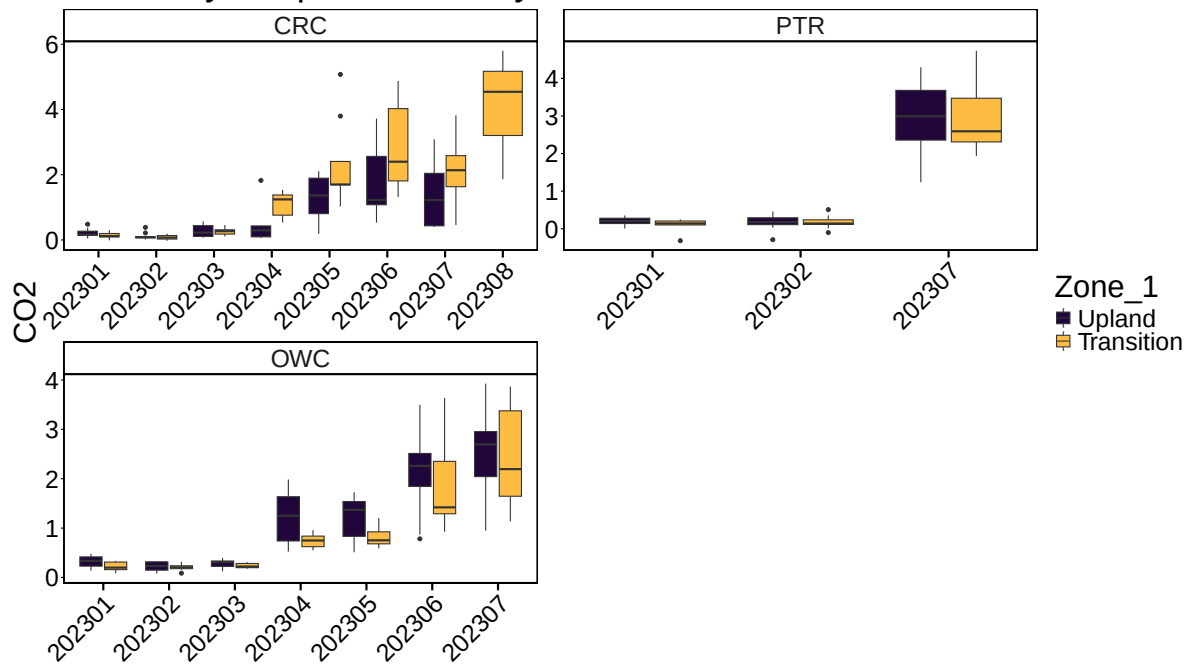




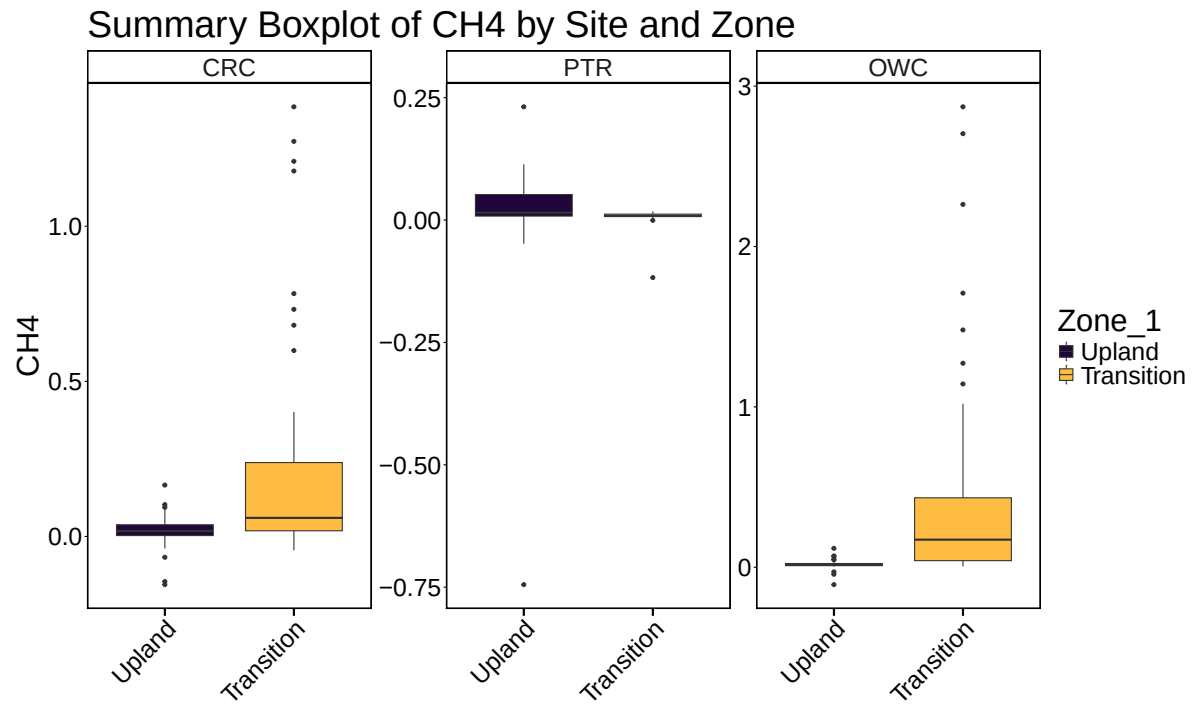
5 Boxplot summary by Date

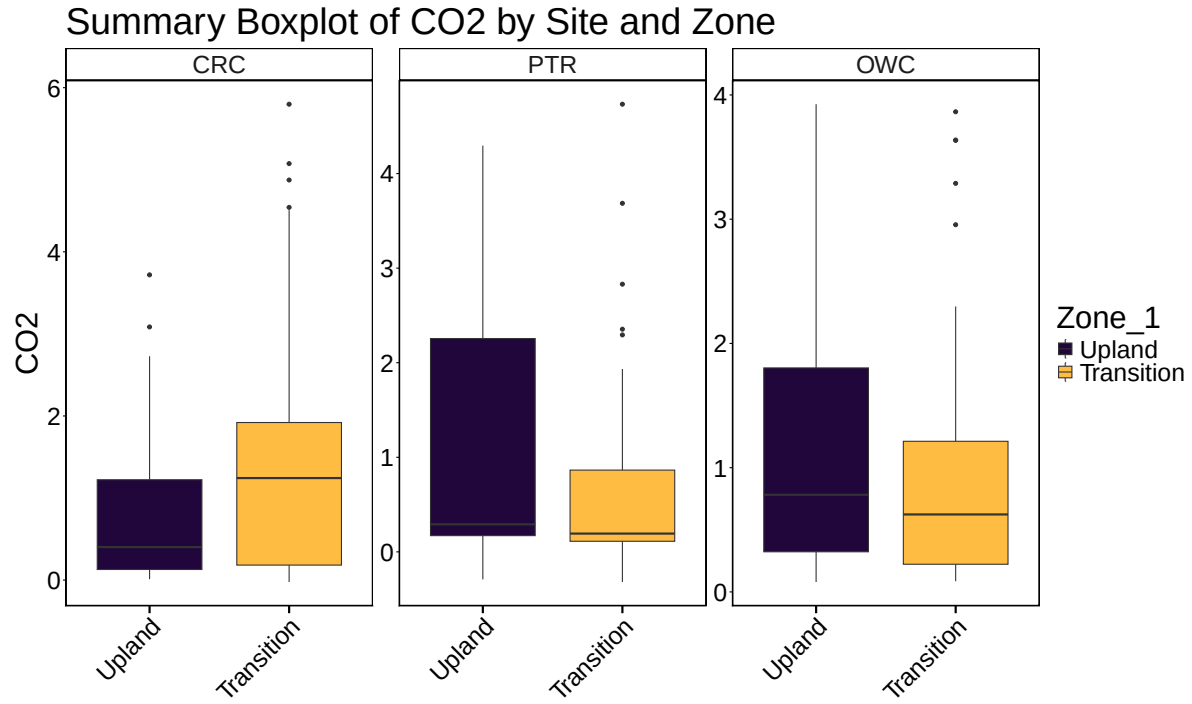


Summary Boxplot of CO2 by Site and Zone



6 Boxplot summary by Site





7 Calculate summary metrics

8 Summary Tables

Table 2: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
CRC	Upland	-0.1552949	0.1656322	0.0183426	0.0116738	3.719250	0.8039148
CRC	Transition	-0.0444538	1.3849897	0.2146893	-0.0232152	5.798788	1.4336332
PTR	Upland	-0.7447805	0.2312600	0.0029840	-0.2913236	4.294703	1.0734774
PTR	Transition	-0.1175338	0.0177527	0.0039678	-0.3186061	4.733226	0.8503890
OWC	Upland	-0.1082825	0.1181312	0.0172978	0.0798020	3.925557	1.1361656
OWC	Transition	0.0055519	2.8710651	0.4297739	0.0862320	3.864481	0.9329563